Shuangqi Li

Lausanne, Switzerland shuangqi.li@epfl.ch lishuangqi.com

About

PhD candidate at EPFL, advised by Mathieu Salzmann. My work spans **both LLM post-training and data-centric pretraining**, focusing on **how training data and learning signals affect model behavior**.

Strong engineering skills in Python, PyTorch, and verl, with hands-on model training and evaluation experience and a competitive programming background.

Education

EPFL (Swiss Federal Institute of Technology in Lausanne) — *Lausanne, Switzerland*
Ph.D. in Machine Learning (expected; timing flexible) — *Sept 2022 – Sept 2027*

EPFL (Swiss Federal Institute of Technology in Lausanne) — *Lausanne, Switzerland*
M.Sc. in Data Science — *Sept 2020 – July 2022*

University of California, San Diego — *Remote*
Graduate studies in Computer Science (not completed due to COVID) — *Sept 2019 – June 2020*

University of Electronic Science and Technology of China — *Chengdu, China*
B.Eng. in Microelectronics Science and Engineering — *Sept 2015 – June 2019*

Research Projects

Utility-Guided Credit Assignment for LLM Post-Training — *June 2026 – present*

Ongoing research. On-policy distillation work under review for ICLR 2027.

- Developed **reward-gradient alignment** to estimate the utility of token-level updates and reweight **teacher supervision in on-policy distillation**, emphasizing beneficial signals and reducing harmful ones.
- Enabling turn-level **dense credit assignment for long-horizon agentic RL** guided by its **gradient utility** on other transferrable tasks.

Influence-Guided Self-Evolving Data Synthesis — *Aug 2026 – present*

Ongoing research.

- Developing an iterative data/environment-generation pipeline that uses gradient-based probes to identify capability gaps, retrieve informative seeds, and **synthesize agentic-SFT trajectories and agentic-RL environments targeting model weaknesses**.

PriFT: Guiding Post-Training with Pretrained Prior — *Feb 2026 – May 2026*

NeurIPS 2026. paper

- Developed **pretrained-prior-aware SFT** to improve generalization on mathematics, coding, and question answering, while providing a **stronger initialization for subsequent RL**.

LoRIF: Pushing the Scalability Frontier of Training Data Attribution — *Aug 2025 – Jan 2026*

NeurIPS 2026. paper

- Developed low-rank influence functions, reducing gradient storage and query latency by up to 20× while matching or improving attribution quality, **enabling data attribution at modern scale**.
- **Scaled attribution to Apertus-70B over 3.8M SFT examples** on two H200s, linking individual responses to influential training data for model diagnosis and auditing.

Learning to Weight Parameters for Training Data Attribution — *Feb 2025 – Sept 2025*

ICLR 2026. paper

- Revealed attribution heterogeneity across model layers and developed self-supervised reweighting to improve attribution across **image generation, classification, and language modeling**.
- Enabled **fine-grained attribution**, identifying influential training examples for different capability targets.

LLM Development from Scratch *Mar 2025 – July 2025*

EPFL EE-628 course project, developed in a team.

- Engineered an **LLM pretraining and evaluation workflow**, explored data mixtures, and reproduced SmolLM2 benchmark evaluations.

Work Experience

Research Intern — LLM Pretraining *Beijing, China*
Ubiquant | IQuestLab — Foundation Team (IQuest-Q1: 320B MoE) *May 2026 – present*

- Analyzed **pretraining corpus quality and diversity** through embedding-based clustering of billion-scale documents; designed **cluster-level sampling and filtering strategies** and evaluated their impact through model training experiments.
- Built a scalable gradient-based attribution pipeline for **capability-targeted training data selection**; validated retrieved examples through **continued pretraining**, improving mathematics and reasoning performance.
- Developed **embedding-based semantic deduplication** following exact and fuzzy deduplication; compared strategies by downstream model performance and token efficiency.

Research Intern *Zurich, Switzerland*
Oracle Labs *July 2021 – Sept 2021*

- Developed a time-series model to detect anomalous Linux sessions on cloud servers, and built PySpark pipelines to process large-scale cloud data.

Algorithm Engineering Intern *Beijing, China*
DiDi *Oct 2018 – Feb 2019*

- Developed an algorithm that learned road-segment weights from historical trip data, improving production route-planning quality for ride-hailing services.

Teaching & Supervision

- **Improving Waste Detection and Sorting** @ *WasteFlow – Project Supervisor* *Fall 2025*
- **CS-233: Introduction to Machine Learning** – *Head Teaching Assistant* *Spring 2025*
- **CS-456: Deep Reinforcement Learning** – *Student Teaching Assistant* *Spring 2022*

Honors & Awards

- **National Scholarship** *Sept 2018*
- **China Collegiate Programming Contest** – Gold Medal *May 2018*
- **ACM ICPC (Asia Regional)** – Bronze Medal *Oct 2017*

Skills

Programming: Python, C++, CUDA; competitive programming

Frameworks & Tools: PyTorch, Docker, Git, Linux, PySpark, verl, Claude Code, Codex

Languages: Chinese (native), English (fluent), French (A1)

Selected Publications

PriFT: Prior-Support Guided Supervised Fine-Tuning *NeurIPS 2026*
Ke Wang[*], *Shuangqi Li*[*], *Mathieu Salzmann, and Pascal Frossard* [*]*Equal contribution*

Low-Rank Influence Functions for Scalable Training Data Attribution *NeurIPS 2026*
Shuangqi Li, Hieu Le, Jingyi Xu, and Mathieu Salzmann

Learning to Weight Parameters for Training Data Attribution *ICLR 2026*
Shuangqi Li, Hieu Le, Jingyi Xu, and Mathieu Salzmann

Enhancing Compositional Text-to-Image Generation with Reliable Random Seeds *ICLR 2025*
Spotlight
Shuangqi Li, Hieu Le, Jingyi Xu, and Mathieu Salzmann

Controlling the Fidelity and Diversity of Deep Generative Models via Pseudo Density *TMLR 2024*
Presented at ICLR 2025
Shuangqi Li, Chen Liu, Tong Zhang, Hieu Le, Sabine Süsstrunk, and Mathieu Salzmann